

Bacteriological Quality and Physicochemical Properties of Drinking Water Sources in Slums of Lagos State, Southwest, Nigeria

Aniakor, Leonard Ubanagu^{1*}; Udeani, Theophilus Kachi²; Ogudu, Emmanuel Ogbonnia³; Ikechi, Amarachi Uche⁴; Eze Chinazo Onyejere⁵

^{1,2}University of Nigeria Enugu Campus, Department of Medical Laboratory Science, Faculty of Health Sciences and Technology, Enugu State, Nigeria.

³National Arbovirus and Vectors Research Centre, Enugu. Federal Ministry of Health, Nigeria.

⁴University of Nigeria Enugu Campus, Laboratory unit, University Medical Centre, Enugu, Nigeria.

⁵Peculiar Laboratory and Diagnostic Centre Ngwo, Enugu, Nigeria

Corresponding Author: Aniakor, Leonard Ubanagu^{1*}

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Abstract: Water is very essential and widely used by humans, animals and for the success of agricultural products. The inhabitants resort to use of borehole, sachet and bottle water for drinking and domestic use. There is paucity of information on bacterial contamination and physicochemical properties of drinking water sources in major slums of Lagos city. Such contaminants may be the cause of frequent febrile illnesses experienced amongst the inhabitants of these slums. The aim of this study is to determine the bacteriological quality and physicochemical properties of drinking water sources in slums of Lagos State, Nigeria. The six selected slum areas are Mushin, Makoko, Bariga, Ijora, Ajegunle and Oniru; the selection was based on the population size of the area. The specific objectives of the study are to (i) determine the bacteria quality of drinking water sources in slums of Lagos State, (ii) determine the total coliform and E.coli count of the borehole water in the slums of Lagos State, and (iii) determine the physicochemical properties of the various sources of drinking water in the different slums of Lagos State. A total of 396 water samples were analyzed for the presence of pathogenic bacterial contaminants using Membrane Filtration Techniques, Culture and Biochemical assays to identify the bacterial isolates. Other appropriate standard techniques were used to determine the physicochemical parameters like pH, temperature, electrical conductivity, turbidity, chloride, nitrate, iron, lead and cadmium in the water samples. A total of 50.8% different bacterial species were obtained from the borehole and packaged waters meant for human consumption. The percentage pathogenic bacteria contaminants were spread across the boreholes, sachet and plastic bottled water as follows 43.7%, 6.6% and 0.5% respectively. The highest bacterial contaminant isolated from borehole water was *Klebsiella pneumoniae* at a frequency of 8.3 % and thermo tolerant *Escherichia coli* 7.3%. The total coliform count in borehole was generally high but highest in Oniru slum while the total *E.coli* count in borehole were higher in Ajegunle than the rest of the slums. Relatively very few bacterial contaminants were isolated from sachet water with *K.pneumoniae* 2.5% still representing the highest while only *E.coli* was isolated from plastic bottled water at a very low percentage 0.5%. This present study has revealed that drinking water sources in the slums of Lagos State are unsafe for drinking due to the presence of pathogenic bacteria, solid minerals and heavy metals beyond the acceptable limits.

Keywords: Slums, Membrane Filtration, *Escherichia Coli*, *Klebsiellae Pneumoniae* Coliforms and Physicochemical.

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I. INTRODUCTION

Water is very essential and widely used by humans, animal and for the success of agricultural products. The sources of water include streams, rivers, sea, ocean and man-

made adaptations. These adaptations include construction of borehole water, digging of wells and holistic water treatment plants managed by the government organs. The use of water by human requires that such water system should be clean and safe from bacteria contaminants so as to avoid water borne

diseases. The target of UN-Sustainable Development Goal No. 6 which is to provide clean water and sanitation for all by the year 2030. This is an uphill task within Lagos State with an increasing population due to influx of migrants both within and outside Nigeria. The provision of adequate and safe potable water to the people in any nation is a responsibility of government both at the federal and state levels but unfortunately in many parts of the developing world, especially in urban slums, these fundamental right remains unrealized. According to the World Health Organization (WHO, 2023), approximately 2.2 billion people globally lack safely managed drinking water. In Lagos, Nigeria with a teeming population of over 20 million such is equally unrealized. About 56% of Lagos residents live in slums according to UN-Habitat (2021) report. The United Nations Human Settlements Program (UN-Habitat, 2020) defines slums as "residential areas where inhabitants lack one or more of the following: durable housing, sufficient living area, access to clean water, access to sanitation, and secure tenure. Arimah and Branch (2018) define slums as "unplanned urban settlements where poverty, poor infrastructure, and weak institutional oversight interact to produce unhealthy living conditions. Most of the slum dwellers are low and middle income earners. Rapid migration and urbanization for better life results to population growth and overcrowding.

As a coastal State, greater parts of Lagos is surrounded by ocean water (Atlantic Ocean) and some these areas that are water logged with Shanty dwelling homes while the hinterlands grapples with over population and poor sanitary conditions. These factors make the supply of water an individual affair which leads to private individuals constructing bore holes and packaging water in cellophane papers and plastic bottles. The bacteria quality of this water may not be well regulated as the regulators may not reach some of the Shanty areas for environmental hygiene and proper piping for such packaging. The bacteria quality, cleanliness and safety of these water systems may not be assured which may lead to intermittent water borne diseases (WHO, 2024).

Many slum households do not have access to toilets or proper sewage systems, which results in open defecation or the use of makeshift pit latrines. The practice of this open defecation and the use of inadequately designed pit latrines contribute to a large extent to the ground water contamination through the infiltration of fecal contaminants in the soil. (Odo *et al.*, 2024).

Another major issue is the prevalence of unregulated and informal water provision. In the absence of a formalized public supply, water is sourced from not monitored boreholes, hand-dug wells, and surface water none of which are subjected to routine quality control or treatment. Informal water vendors sell water of questionable origin and quality, exposing residents to further risk. Compounding this is the issue of poor solid waste management. In many slum communities, garbage is disposed of in open spaces or drains, allowing pollutants to leach into the soil and groundwater systems (Dauda *et al.*, 2024; Lu *et al.*, 2025; Alao *et al.*, 2025).

Additionally, the topographical and geographical location of many slums contributes to the risk of contamination. These communities are often situated in low-lying, flood-prone zones or near industrial zones where effluents are discharged without regulation. During the rainy season, water from contaminated surfaces and sewage channels floods into wells and open water sources, making them unsafe. Finally, weak governance and inadequate policy implementation further exacerbate the problem. Bello *et al.*, (2021) observe that urban planning efforts in Lagos rarely prioritize informal settlements, resulting in a regulatory vacuum that leaves slum dwellers without oversight or protection from unsafe environmental conditions.

The public health implications of contaminated drinking water in slum communities are extensive and severe. One of the most immediate and common consequences is the outbreak of waterborne diseases. Diarrhea diseases, for instance, are rampant in these environments and remain one of the leading causes of child mortality, particularly among children under the age of five. The ingestion of fecal-contaminated water also leads to the spread of infectious diseases such as cholera and typhoid, which thrive in overcrowded environments where clean water is scarce and sanitation is inadequate (WHO, 2024).

Cholera, in particular, poses a serious risk in slum settings due to the rapidity with which it spreads through contaminated water sources and the lack of immediate medical attention. Hepatitis A is another disease frequently associated with the prolonged use of unsafe drinking water, leading to liver complications in both children and adults. Skin and eye infections are common among residents who bathe or wash with untreated water, especially in areas with stagnant or polluted surface water. Beyond the immediate health threats, long-term exposure to contaminated water has profound developmental implications especially on children (UNICEF, 2024).

Chronic exposure to microbial and chemical contaminants in drinking water can lead to stunted growth, malnutrition, cognitive impairment and weakened immune systems in children. These health effects perpetuate cycles of poverty and hinder educational attainment, especially in already disadvantaged slum populations (Shrestha *et al.*, 2021; and WHO, 2023; WHO, 2024)

Lagos State, the most populous urban city in Nigeria, has witnessed rapid urban expansion over the past decades. However, much of this growth has been unregulated, resulting in the proliferation of extensive slum settlements

The goal of this study is to investigate the bacteria quality of borehole, sachet and package waters in selected slums in different local government areas of Lagos state. This is to evaluate the target of UN-SDG.6 goal in assuring the continued maintenance of clean and safe water for the populace well-being.

➤ Aim of the Study

The aim of this study is to determine the bacteriological quality and physicochemical properties of drinking water sources in slums of Lagos State, Nigeria.

➤ The Specific Objectives of the Study are to

- Determine the bacteria quality of drinking water sources in slums of Lagos State,
- Determine the total coliform and E.coli count of the borehole water in the slums of Lagos State, and

- Determine the physicochemical properties of the various sources of drinking water in the different slums of Lagos State

II. MATERIALS AND METHODS

➤ Study Area

The study was done in six Local Government Areas of Lagos State, Nigeria. Six major slums were selected for this research namely Makoko, Ajegunle, Bariga, Mushin, Ijora and Oniru based on their high population density. Lagos State coordinates are Latitude: 6.23° N-6.70°N. Longitude: 2.70° - 4.20° E.

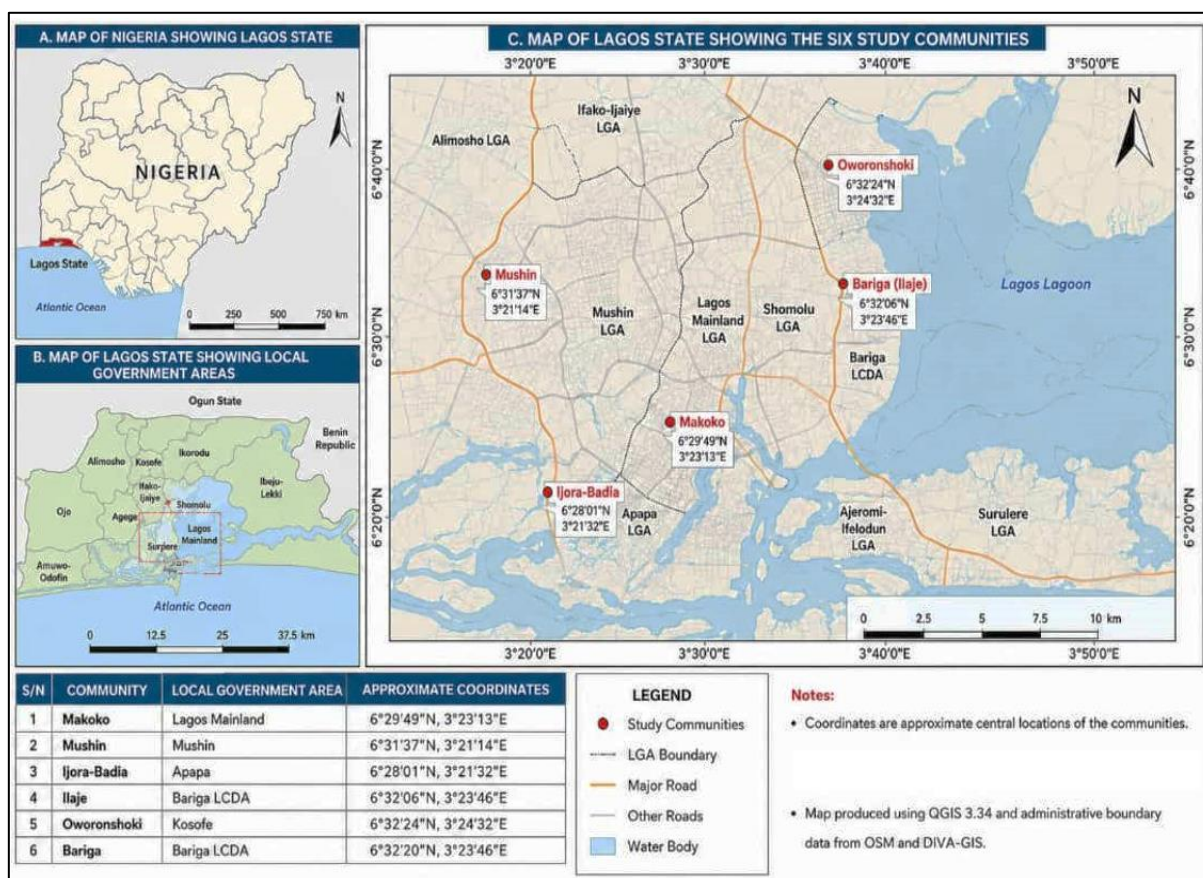


Fig 1 Study Area

➤ Study Design

A cross sectional study was used to determine the microbial quality and physiochemical properties of drinking water sources [borehole, sachet water and bottled water] in the slums under study. Global Positioning System (GPS) was used for accurate location of the slums and water samples distribution

➤ Sample Size and Sample Collection

A total of 396 water samples were collected from different locations including, 226 borehole, 85 sachet water and 85 bottle water. Water samples were collected aseptically in duplicates with a 500 ml sterile plastic bags and 200ml sterile bottle according to the standard procedure stated in WHO drinking water guideline (WHO 2024). For each 500ml sample collected, it was sent immediately to Yabatech

Molecular Central laboratory with an ice box for microbiological analysis while the 200ml was sent for physico-chemical properties at D. K. Olukoya Central Research and Reference Laboratories, University of Lagos, Akoka. Molecular studies of the isolates were carried out at Nigerian Institute of Medical Research (NIMR) Yaba. The sample bottles were clearly labeled with essential information such as the date, time, location, and sample identification number. Immediately after collection, the samples are placed in a cooler box containing ice packs and transported to the laboratory at a temperature of approximately 4°C. Analysis was carried within six hours after collection. Freezing of samples was avoided, as it may destroy bacterial cells. Field blanks (sterile water samples exposed to the sampling environment) were included to monitor possible contamination during collection and transportation.

III. SAMPLE ANALYSIS

➤ Membrane Filtration Technique

Using sterile forceps, a membrane filter with a pore size of 0.45 µm was placed onto the filter holder, ensuring that the grid side was facing upward. The funnel was then attached securely to the holder. A 100 ml volume of the water sample was poured into the funnel, and gentle vacuum suction applied to draw the sample through the membrane. As the water passed through, bacteria were retained on the membrane surface. After complete filtration the funnel was rinsed with 10–20 ml of sterile distilled water to wash down any residual bacteria onto the membrane. The vacuum was released and the membrane filter was carefully removed using sterile forceps. The membrane was placed, bacteria side up, onto the surface of a prepared and labeled selective agar plate EMBA and McConkey agar. Care was taken to ensure that no air bubbles were trapped between the filter and the agar surface, as this may hinder bacterial growth. The plate was then incubated under conditions appropriate for the target organisms: for example McConkey agar incubated at 35°C for 18–24 hours to detect total coliforms, while EMBA are incubated at 44.5°C for 24 hours for fecal coliforms especially the thermotolerant *Escherichia coli*. Following incubation, visible colonies develop on the membrane surface, representing bacteria originally present in the sample. The results are later interpreted as colony-forming units per 100 ml of water (Aziz, *et. al.*, 2024).

➤ Coliform /*E.coli* Counts.

Using electronic automatic J.S colony counter, the coliforms and *E.coli* from McConkey agar and EMB agar were counted and recorded as CFU/100ml. Cultures yielding colonies with a green metallic sheen typical of *Escherichia coli* on EMB agar plates were counted and recorded too. The colonies from the McConkey and EMB agar plates were re-streaked repeatedly for three (3) times to obtain a pure isolate.

➤ Biochemical Assays

Further biochemical tests were performed on the bacterial isolates to identify the organisms at least to generic level. The biochemical assays performed on the isolates include, Catalase, Coagulase, Urease, Indole, Oxidase and Citrate utilization tests. Identification of individual isolate was based on colonial characteristics, Gram reaction and biochemical tests

IV. RESULTS AND DISCUSSION

Water is an indispensable natural resource essential for human survival, and access to microbiologically safe drinking water is fundamental to the prevention of waterborne diseases. The World Health Organization (WHO, 2024) recommends that drinking water intended for human consumption should be free from faecal indicator bacteria and pathogenic microorganisms because their presence poses significant risk to public health.

In the present study, 50.8% of the bacterial isolates recovered from borehole, sachet and bottled water is indicative of substantial microbial contamination and

constitutes a major public concern. The isolation of *Klebsiella pneumoniae* (8.3%), *Escherichia coli* (7.3%), *Salmonella typhi* (2.3%), *Salmonella paratyphi* (1.3%), *Pseudomonas aeruginosa* (0.8%), *Enterococcus faecalis* (3.8%), and *Proteus vulgaris* (1.0%) demonstrates that a considerable proportion of the water sources investigated had been compromised by faecal contamination, environmental pollution, inadequate sanitation, poor handling practices, or deficiencies in water treatment and distribution systems. The detection of *E.coli* is particularly significant because it is a universally accepted indicator of recent faecal contamination and suggests the possible presence of other enteric pathogens.

The occurrence of these bacteria species in borehole, sachet and bottled water is of public health importance because they are associated with gastroenteritis, typhoid fever, urinary tract infections, septicemia, wound infections and opportunistic infections, particularly among children, older adults and immunocompromised individuals. Therefore. The presence of these organisms in drinking water suggests that some of the sampled water sources may not be microbiologically safe for human consumption without appropriate treatment. These findings underscore the need for improved water source protection, routine microbiological surveillance, effective treatment and strict compliance with the World Health Organization and Nigerian Standard for Drinking Water Quality (NSDWQ) guidelines to reduce the burden of waterborne diseases.

The bacteria isolates of 50.8% in this study were spread across the bore holes, cellophane and plastic water as follow 43.7%, 6.6% and 0.5% respectively. The boreholes water serves as a major sources of water for domestic use and serve the water packaging industry. Ape *et al.*, (2025) likewise reported high bacteria pathogens such as *E.coli*, *Klebsiella*, *Salmonella*, *Proteus*, *Enterobacter*, *Bacillus* and *Staphylococcus species* associating the heavy contamination with fecal contamination. A report from Yahaya *et al.*, (2020) reported high isolation bacterial isolates at Iwaya, Makako and Ilaje. Comparable studies from Oyo, Osun and Enugu States have consistently reported high bacterial contamination with waterborne pathogens (Okpasuo *et al.*, 2020) in Enugu State. These findings suggest that microbial contamination of groundwater and household water supplies is a widespread public health concern in Nigeria, particularly with inadequate sanitation and poor environmental hygiene, this shows that Lagos findings are consistent with reports from other regions of Nigeria.

In drinking water processing industries, the systemic production line from boiling to filtration may have reduced bacterial contaminants in sachet and bottled water (Ghislain *et al.*, 2010; Agrawal *et al.*, 2011, Zha *et al.*, 2025) reported high reduction of bacteria load through boiling, filtration and solar disinfection. The coliform count as obtained from borehole water indicated high numbers of bacteria presence in Oniru, Ajegunle, Ijora and Mushin with mean of 108.5 ± 33.47 cfu/100ml, 80.28 ± 20.01 CFU/100ml, 72.66 ± 39.96 CFU/100ml and 69.02 ± 41.06 CFU/100ml respectively, thus exceeding the limit set by World Health Organization

guidelines for drinking water quality 2022 and Nigerian Standard for drinking Water quality 2020 set by Nigeria Industrial Standard (NIS). This in agreement with Ojo *et al.*, (2021) that reported in Ilaro, Southwest Nigeria high bacterial and coliform counts in borehole. In contrary, Adeyemi, (2020) reported absence of coliforms from borehole waters in Olusegun Agagu University of Science and Technology Okitipupa, Nigeria. He concluded that the borehole water was clean for human drinking.

The mean coliform count were performed differently for both coliform and *E. coli*. The count indicated that bore hole waters from Oniru, Ajegunle, Ijora and Mushin had mean bacteria count of 108.5 $\pm$ 33.47, 80.28 $\pm$ 29.91, 72.66 $\pm$ 39.96 and 69.0 $\pm$ 41.06 respectively. In borehole water from Bariga and Makoko showed mean bacteria count showed mean bacteria count of 41.94 $\pm$ 21.67 and 67.87 $\pm$ 52.21.

The *E.coli* ranges from 2.33 $\pm$ 1.29cfu/100ml in Ajegunle and 1.80 $\pm$ 0.92cfu/100ml in Makoko and other Local Government Areas respectively have far above acceptable limits as recommended by World Health Organization (WHO, 2022; NSDWQ, 2020). Adekanmi *et al.*, (2020) in Ibadan reported high levels of coliform and *E. coli* in wells. Lawal *et al.*, (2024) indicated that contamination of borehole water may have been sewage seepage and open defaecation around the borehole site. In similar studies, (Adebayo *et al.*, 2021; Adekunle and Oyebanji, 2021) reported *E. coli* due to fecal contamination from infiltration. Hinton *et al.*, (2024) surveyed sanitation infrastructure in Malawi slums and identified that closer proximity to latrines correlated with elevated coliform counts. Distance of 30 meters (100 feet) is recommendable Murat Ersel, (2016).

Water borne diseases are majorly due to presence of bacteria in water systems for domestic uses. It can be as a result of inherent presence of these bacteria agents or storage tanks. Personal and environmental hygiene may also alter the constituents of water for domestic use due to arbitrary introduction of bacterial agents. Internally people assume that borehole water, sachet water and bottled water are free from contaminating microorganisms. This assumption is misleading as drinking waters are prone to bacterial habitation.

In this study, the predominant bacteria isolate from borehole and sachet waters were *Klebsiella pneumoniae*. This is in agreement with (Mba and Amuche, 2024; and Adomi *et al.*, 2025) who reported predominance of *Klebsiella species* in Elele, Rivers and Delta State respectively. However, this report differs from previous reports that mostly reported the predominance of *E.coli* in these water systems (Olalemi *et al.*, 2021; Okere *et al.*, 2021; Okoye and Nyandjou, 2024), . The *Salmonella species*, *Enterococcus faecalis* and *E.coli* are typical indicators of bacterial contamination of water. The presence of *thermo tolerant E. coli* and aerobic *E. coli* in this study indicated human or animal fecal contamination. Adeyemi *et al.*, (2019) in Osogbo reported high *E.coli* in well water close to septic tanks. The only bacterium isolate from

bottled water was *E. coli* which may be due to poor personal hygiene of the workers at the point of bottling and corking (Maselela *et al.*, 2024).

The importance of thermo tolerant *E.coli* as obtained in bore hole and sachet water indicated persistence of this bacterium in these water bodies. This bacterium (its presence) in bore hole water constitute a serious risk to human health as the organism may induce disease of unknown origin especially in children, elderly and immunocompromised host. This is because the thermo tolerant *E. coli* in trying to adapt in human stomach region, will compete with normal flora in the stomach, thereby secreting toxins that may induce a disease condition. Despite contacting this thermo tolerant *E.coli* from water, it can be contracted through vegetables, street foods and poorly cooked meat. Luna *et al.*, (2019) reported how raw vegetable can be a source of pathogenic *E. coli* transmission at any point from pre- to post harvest. In a similar study Farrokh *et al.*, (2013) reported shigella-toxing producing *E. coli* in vegetable meant for human consumption.

The rampant episodes of Salmonellosis in Nigeria may be attributed to consumption of poorly treated water systems. The detection of *Salmonella species* in this study from borehole and sachet waters indicated that the water systems may be a major vehicle for this disease (Yusuf *et al.*, 2021). Akinyemi *et al.*, (2011) reported 18.5 % *Salmonella species* of 200 water samples (well, pipe borne and sachet water) from Lagos communities including Ajegunle, Mushin, Bariga, Oshodi, Agege, Ikorodu and Badagry. Sila, (2018) in Kenya reported high presence of *Salmonella species* from river water while its absence was observed in borehole water. The difference in this report about borehole water may be the location of the borehole far from human/animal defaecation point.

Another important pathogen isolated was *Pseudomonas aeruginosa* at a frequency of 3.0% from borehole water only. In other Nigerian studies, apart from Emmanuel- Akerele *et al.*, 2021 who recovered *P. aeruginosa* in borehole in Ayobo, Lagos, prevalence range from 16% of bacterial isolates in Abakiliki boreholes to 49% occurrence in Uzuakoli boreholes while a recent investigation in Katsina detected *P. aeruginosa* 71.1% of drinking water samples with borehole among the most contaminated sources. It's presence in borehole water may represent poor sanitation practices and biofilm formation on the piping system. The occurrence of *Pseudomonas aeruginosa* in borehole water is an indicator that the water system supports the growth and survival of the microbe because this bacterium is an opportunistic environmental organism that can survive in adverse condition. Bengani *et al.*, (2025) reported on *Pseudomonas aeruginosa*'s diversity and abundance in drinking water. Erika *et al.*, (2018) isolated *Pseudomonas aeruginosa* and Ohalet *et al.*, (2025) reported presence of *P. aeruginosa* in borehole water as a public health risk in Owerri.

Of importance, the mixed bacteria growth from borehole showed that such boreholes that such boreholes are grossly contaminated. In some boreholes, mixed bacteria isolates were observed and this includes *E.coli/ Pseudomonas*

aeruginosa/ Klebsiella pneumoniae and *E.coli/Klebsiella pneumoniae/Pseudomonas aeruginosa* are indicators of biofilm formation in the piping system of the borehole waters due to poor maintenance of the facilities of the borehole. Secondly, in many Lagos slums, boreholes are often shallow and inadequately protected, making them susceptible to bacterial contamination from nearby open drains, canals and floodwaters during rainy season. (Yahaya *et al.*, 2021). Okoye *et al.*, (2023) in Abeokuta reported more contamination in low lying wells and Raphael *et al.*, (2023) reported due to flooding in Akwa Ibom State, Nigeria.

The categorization of bacterial isolates in accordance to Local Government Areas indicated that Ajegunle, Mushin and Bariga were the most affected areas with high bacterial contaminants within the boreholes and this relatively affected the sachet water sold in those areas. Yahaya *et al.*, (2021) investigated borehole and well water in Makoko, Iwaja and Ilaje and reported also abnormally high bacterial and coliform counts in borehole water. Akerele *et al.*, (2023) also assessed borehole water quality in Lekki, Lagos and found bacterial contamination, emphasizing the need for routine monitoring and treatment before consumption. The other areas, Makoko, Ijora and Oniru were equally affected by bacterial contaminants of the borehole but not as high as in other areas. Collectively, thermo tolerant *E.coli* and *Klebsiella, Pseudomonas aeruginosa* and *Salmonella typhi* were observed in all the borehole waters collected in all the Local Government Areas sampled. This indicated a pattern of colonization in those areas. This pattern consisted of poor hygienic practices and or a mixture of water drainage from the surrounding dirty water canals or flooding draining into those places. (Genter *et al.*, 2021). Emmanuel- Akerele *et al.*, (2021) reported also that borehole water in Ayobo, Lagos contained *Escherichia coli*, *Pseudomonas aeruginosa*, *Salmonella paratyphi*, *Shigella flexneri*, *Proteus vulgaris* and *Citrobacter freundii*, with several isolates exhibiting multidrug resistance. However, Adeyemo, (2020) reported absence of bacterial contamination in borehole water in Olusegun Agagu University of Science and Technology, Okitipupa in Ondo State. He attributed it to adequate clean environment.

Another important factor is that these slum areas studied harbour more of poor economic earners which make them more vulnerable to bacterial contamination of water system due to poor environmental management such as waste cleaning and waste dumps causing infiltration of leechchates containing microorganism, nitrate and heavy metals. (Obunwo *et al.*, 2022; Yusuf *et al.*, 2021 Okafor and Olowu, 2021).

Borehole waters are recognized by their colorless nature and lack of odor. In this study the indicators of water contamination were compared with bacteria isolates. It was observed that 15.9% of the borehole waters that had colour change harbored bacteria agents. The most worrisome is the borehole waters without any tinge of colour that harbored 27.8% bacterial agents. This constitutes the worst problems in borehole waters in that such water system look neat and colorless and people tend to drink it directly from the

borehole without prior treatment like boiling or domestic filtration. In all, colour as an indicator is very vital to rule out possible waterborne disease that may show different colour water. (Husband and Boxall, (2011).

Human defaecation near the borehole, open drainage and sewage seepage can seriously affect ground water quality and public health by a way of contaminants infiltrating the soil and enter the aquifer supplying the borehole, thereby increasing the chances of diarrhoea, cholera, dysentery, typhoid gastroenteritis (Sridhar and Parimalerenganayaki, 2024; WHO, 2024).

In this study, physico-chemical properties are determined in borehole, sachet and plastic bottled water. Generally, low pH values were obtained from Mushin, Bariga, and Makoko at the rate of 12% each while Ajegunle recorded 4% and Ijora 8% (low pH values of 5.52 ± 0.72). It was observed that in the five Local Government Areas, some of the borehole had acidic pH. The impact of these conditions is that it will have corrosive effect on the water piping system, thereby inducing water contamination. Some of the borehole waters are within the normal range of World Health Organization and Nigerian standard for drinking water quality (NSDWQ) of 6.5 and 8.5. This variation on pH attack on piping may allow the formation of biofilm by some of the bacteria, hence the high number of bacterial agents. Earlier studies by Boone and Xun, (1987) reported that pH 7.0 is vital for growth and reproduction of the majority of mesophilic pathogenic bacteria involved in the bio degradation of organic matters dissolved in water- thus they may be competitive bacterial agent in the water system, those can grow at acidic and neutral pH. Therefore, this indicate poor maintenance of the water system as the piping may require changing never been implemented. Akinbile and Ogunniyi, (2023) reported in Lagos slums levels outside this safe range.

In this study, borehole with mean value neutral pH 7.03 ± 0.37 supports survival of many bacteria, improves nutrient availability and promotes biofilm production. Iorkpili *et al.*, (2025) in Benue reported that *E.coli* viability and transport in groundwater movements was seamless..

The acidic pH value noticed in the borehole water may be the reason for the persistent thermo tolerant *E. coli* because they may have acquired mutation due to selective pressure or co-selection (Seiler and Berendonk 2012). Gallo *et al.*, (2025) reported stress adaptation leading to mutation of *E.coli*.

Electrical conductivity determines the capability of water to allow electric current to pass through. Since the electrical conductivity is conducted through the movement of the ions in fluid form, the electrical conductivity also gives a clue of total dissolved salts (TDS) in the water being tested for electrical conductivity. It was observed that electrical conductivity ranges from 439.81 ± 235.28 to $11400.60 \pm 261.71 \mu\text{S/cm}$. The borehole water from the six local government areas indicated low electrical conductivity while 2%, 2% and 16% of borehole water from Mushin, Bariga and

Ajgunle have high electrical conductivity. It has been observed Sila (2019) that pollution related increase in electrical conductivity of water bodies renders such water unsuitable for human use. Okafor *et al.*, (2024) reported electrical conductivity values exceeding 1500 $\mu\text{S}/\text{cm}$ in Lagos slums.

It has been postulated that electrical conductivity decreases during the rainy season due to water dilutions while high electrical conductivity values may render the water unsuitable for human use. The major difference occurs during heavy rainy season while in dry season, there may be drying of the soil and extreme heat which will increase the conductivity of the water system, therefore, treatment of borehole water before domestic use is necessary avoid the public health impact of electrical conductivity because it indicates elevated concentrations of ion salts or minerals in the water. It may indicate contamination from sewage intrusion, industrial discharge, agricultural runoff, leaches from waste dumps and seepage from septic tanks. (Manta *et al.*, (2024). Electrical conductivity is closely related to total dissolved solids in that it can affect taste of the water and cause gastrointestinal irritation in some individuals and lead to hypertension risk from high sodium. Although high electrical conductivity does not increase bacterial load but high electrical conductivity from sewage contamination correlates with the presence of pathogen such as *E. coli*. Enerijiofi *et al.*, (2018) reported increase in coliform and presence of pathogens. Another fact is that the elevated electrical conductivity may indicate filtration of contaminated surface water and poor borehole construction. (Ugbenyen and Godwin, (2022) in Edo State observed seepage and environmental pollution surrounding the borehole. Etesin *et al.*, (2026) in Port Harcourt reported variations in electrical conductivity alongside total coliform contamination.

One of the heavy metals encountered in water is lead. This heavy metal is encountered in water through batteries, old pipes, industrial waste and paints. This metal can find its way to the borehole water due to poor waste control management in Nigeria because either the components of the batteries or other electronic are just dumped anywhere there is space so that its contents will leak into the soil and drain into the borehole water. In this study, high level contents of lead were observed in 4 boreholes namely Ajegunle, Ijora, Bariga and Mushin. This heavy metal is of public health concern because of its renal, cardiovascular, neurological and developmental implications. In recent studies, Ojiodu *et al.*, (2026) reported high levels of lead and cadmium in Ijora-Badia.

The presence of Cadmium particles from borehole waters in Mushin, Bariga, Ajegunle and Ijora is of significance because this heavy metal have been implicated in reduced renal function, interference with calcium metabolism which leads to bone weakness, osteomalacia, osteoporosis and increased risk of bone fracture (He and Zhang, 2025; Castañeda *et al.*, 2025). It has also been shown that presence of cadmium in water increase the risk of

cardiovascular disease, impair growth and development in children and fetal toxicity in pregnant women; thus the occurrence of cadmium in borehole water is of public health concern because of its damaging effects in human.

In this study, the iron content in the borehole waters indicated that 48% of the borehole water collected had high iron content. Its presence is indicative of diverse sources of iron particles within the environment, for instance the welding processes, automobile (mechanics), construction sites, all release iron contents to the environment which finds its way to the borehole waters. The public health implication of iron particles in drinking water includes gastrointestinal irritation, nausea/vomiting in sensitive individuals and risk of iron overload in individual with a disease condition such as hemochromatosis. Generally, iron may not induce major toxic damage but may increase disease condition in immunocompromised individuals (Moro, 2025).

Of importance is the presence of copper and nitrate in the water sources. Copper which is a heavy metal naturally is present in water either as a soil deposit or an indication of pipe corrosion. (Garlarce *et al.*, 2020). It has been shown that copper has bactericidal properties against cholera carrying bacteria and biofilms, thus acting as a deep detoxifying agent against microorganisms. (Asaad *et al.*, 2025). However excess copper in water alters the taste of water to either bitter or metallic taste or an indicator of liver and/or kidney damage when present in high concentration. (Zhou *et al.*, 2025).thus consistent, regular monitoring of copper in water bodies will help to checkmate microbial growth in such water system. The excessive nitrate occurrence water is an indicator of fecal or environmental contamination which is suggestive of sewage or animal waste origin. In this study the nitrate concentrations were of normal values which is in contrast with Umeh and Chukwuma (2025) who reported that reported nitrate level exceeding 50mg/L in over 40% of tested samples.

V. CONCLUSION

This present study has revealed that drinking water sources in the slums of Lagos State are unsafe for drinking due to the presence of pathogenic bacteria beyond the acceptable limit. It is also established that majority of the health related changes facing the inhabitants of these slums could be as a result of drinking bacteria contaminated water. Moreso, the presence of some solid minerals and heavy metals above the acceptable limits in the water sources further made them unsafe for human consumption as it poses health threats ranging from mild febrile conditions to a more complicated cases like organ damage.

Table 1 Summary of Bacteria Isolates from Water Sources

Bacterial isolate	Type of Water Source			
	Borehole	Sachet	Bottled	Total (%)
<i>E. coli</i> (Thermotolerant 44.5°C)	29 (7.3)	4 (1.0)	0	33 (8.3)
<i>E.coli</i> (37°C)	8 (2.0)	2 (0.5)	2 (0.5)	12 (3.0)
<i>Klebsiella pneumoniae</i>	33 (8.3)	10 (2.5)	0	43 (10.8)
<i>Pseudomonas aeruginosa</i>	12 (3.0)	0	0	12 (3.0)
<i>Salmonella typhi</i>	11 (2.8)	-	-	11 (2.8)
<i>Salmonella paratyphi</i>	5 (1.3)	4 (1.0)	0	9 (2.3)
<i>Proteus mirabilis</i>	1 (0.25)	1 (0.25)	0	2 (0.5)
<i>Enterococcus faecalis</i>	15 (3.8)	0	0	15 (3.9)

➤ Mixed Isolates:

<i>E.coli</i> (44.5°C) + <i>Kleb. pneumoniae</i>	17 (4.3)	2 (0.5)	0	19 (4.8)
<i>E.coli</i> (37°C) + <i>Pseudomonas aeruginosa</i>	15 (3.8)	0	0	15 (3.9)
<i>E.coli</i> + <i>Proteus mirabilis</i>	3 (0.8)	0	0	3 (0.8)
<i>Klebsiella pneumonia</i> + <i>Salmonella typhi</i>	11(2.8)	0	0	11 (2.8)
<i>Pseudomonas aeruginosa</i> + <i>Salmonella typhi</i>	3 (0.8)	0	0	3 (0.8)
<i>E.coli</i> + <i>Kleb. pneumonia</i> + <i>Pseudo.aeruginosa</i>	3 (0.8)	3 (0.8)	0	6 (1.5)
<i>Enterococcus faecalis</i> + <i>E.coli</i>	7 (1.8)	0	0	7 (1.8)
Total	173 (43.7)	26 (6.6)	2 (0.5)	201 (50.8)

Table 2 Summary of Bacterial Distribution According to the Locations, Borehole

Bacteria isolate	Ajgunle	Bariga	Ijora	Makoko	Mushin	Oniru	Total
<i>Escherichia coli</i> (44.5°C)	9 (2.3)	5 (1.3)	4 (1.0)	3 (0.75)	6 (1.52)	2 (0.25)	29 (7.3)
<i>Escherichia coli</i> (37.°C)	4 (1.0)	1 (0.25)	0	0	3 (0.75)	0	8 (2.0)
<i>Klebsiella pneumonia</i>	11 (2.78)	6 (1.5)	3 (0.75)	2 (0.5)	9 (2.3)	2 (0.25)	33 (8.3)
<i>Pseudomonas aeruginosa</i>	3 (0.75)	3 (0.75)	1 (0.25)	3 (0.75)	1 (0.25)	1 (0.25)	12 (3.0)
<i>Salmonella typhi</i>	5 (1.26)	1 (0.25)	2 (0.25)	1 (0.25)	1 ((0.25)	1 (0.25)	11(2.78)

<i>Proteus mirabilis</i>	1 (0.25)	0	0	0	0	0	1 (0.25)
<i>Enterococcus faecalis</i>	4 (4.0)	5 (1.26)	1 (0.25)	2 (0.50)	2 (0.50)	1 (0.25)	15 (3.78)
Total	37 (9.34)	21 (5.30)	11(2.78)	11 (2.78)	27 (6.81)	7 (1.8)	114 (28.79)
Mixed isolates							
<i>Escherichia coli</i> (44.5°C)	6 (1.5)	3 (0.75)	1 (0.25)	2 (0.50)	5 (1.26)	0	17 (4.29)
<i>Escherichia coli</i> + <i>P.pneumonia</i> (37°C)	4 (1.00)	3 (0.75)	2 (0.50)	2(0.50)	3 (0.75)	1 (0.25)	15 (3.79)
<i>E.coli</i> + <i>Proteus mirabilis</i>	1 (0.25)	0	0	1 (0.25)	1 (0.25)	0	3 (0.75)
<i>pneumonia</i> + <i>Salmonella typhi</i>	3 (0.75)	2 (0.5)	1 (0.25)	2 (0.50)	3 (0.75)	0	11 (2.78)
<i>P.aeruginosa</i> + <i>Salmonella typhi</i>	1(0.25)	0	0	2 (0.50)	0	0	3 (0.75)
<i>E.coli</i> + <i>K. pneumonia</i> + <i>P. aeruginosa</i>	0	1 (0.25)	0	1 (0.25)	1 (0.25)	0	3 (0.75)
<i>Faecalis</i> + <i>E.coli</i>	2 (0.50)	1 (0.25)	1 (0.25)	1 (0.25)	2 (0.50)	0	7 (1.75)
Total	17 4.29)	10 (2.53)	5 (1.26)	11 (2.78)	14 (3.54)	1 (0.25)	

Table 3 Distribution of Bacteria Isolates from Water Collected from Various Local Government Areas: Sachet Water and Bottled Water

Bacteria isolate	Ajgunle	Bariga	Ijora	Makoko	Mushin	Oniru	Total
Sachet water	(%)	(%)	(%)	(%)	(%)	(%)	
<i>Escherichia coli</i> (44.5°C)	1 (0.25)	2 (0.5)	0	1(0.25)	1(0.25)	0	4 (1.0)
<i>Escherichia coli</i> (37.0°C)	0	0	0	0	2 (0.50)	0	2 (0.50)
<i>Klebsiella pneumoniae</i>	4 (1.0)	1 (0.25)	2 (0.50)	1(0.25)	1 (0.25)	1(0.25)	10 (2.50)
<i>Salmonella paratyphi</i>	1 (0.25)	1 (0.25)	0	0	2 (0.50)	0	4 (1.0)
<i>Proteus mirabilis</i>	0	0	0	0	0	1 (0.25)	1 (0.25)
Mixed Growth							
<i>Escherichia coli</i> + <i>Klebsiella pneumoniae</i>	0	0	1(0.25)	0	1 (0.25)	0	2 (0.50)
<i>E coli</i> + <i>Kleb. Pneumoniae</i> + <i>P. aeruginosa</i>	1 (0.25)	1 (0.25)	0	0	1 (0.25)	0	3 (0.75)
Total	7 (1.78)	5 (1.3)	3 (0.8)	4 (1.0)	5 (1.3)	2(0.5)	26
Bottled water							
<i>Escherichia coli</i> (37.0°C)	0	1(0.25)	1 (0.25)	0	0	0	2 (0.50)
Total	0	1(0.25)	1 (0.25)	0	0	0	2 (0.50)

Table 4 Mean Total Coliform Count in Borehole Water Samples.

Locations	N	Mean total count (Cfu/100ml)
Mushin	41	69.02 ± 41.06
Bariga	34	41.94 ± 21.67
Ajgunle	25	80.28 ± 29.01
Ijora	36	72.66 ± 39.96
Makoko	31	67.87 ± 52.21
Oniru	6	108.5 ± 33.47

Table 5 Mean Total *E. coli* Count in Borehole Water Samples

Location	N	Mean total count Cfu/100ml
Mushin	11	1.64 ± 0.67
Bariga	9	1.56 ± 0.73
Ajgunle	15	2.33 ± 1.29
Ijora	10	1.80 ± 0.92
Makoko	11	1.09 ± 0.30
Oniru	2	5.5± 0.71

Table 6 Physicochemical Properties of Sachet and Bottled Water Samples

Parameter	N	Sachet M±SD	N	Bottle M±SD
PH				
Normal	20	7.63±0.54(55.6)	16	7.65±0.52(44.4)

Temp				
Low	20	23.16±0.85(55.6)	16	23.77±0.83(44.4)
E.Conductivity (µs/cm)				
Low	20	40.25±22.27(55.6)	16	60.88±2.99(44.4)
TDS (mg/dL)				
Normal	20	24.95±10.62(55.6)	16	28.94±3.21(44.4)
CL- (mg/dL)				
Low	20	27.24±6.89(55.6)	16	43.25±3.63(44.4)
NO ₃ ⁻ (mg/dL)				
Low	20	0.31±0.17(55.6)	16	1.08±0.09(44.4)
Cu (mg/dL)				
Normal	20	0.13±0.04(55.6)	16	0.01±0.01(44.4)
Fe (mg/dL)				
Low	15	0.14±0.02(41.7)	7	0.18±0.11(19.4)
High	5	0.56±0.14(13.9)	2	0.30±0.00(5.66)
Pb (mg/dL)			7	0.68±0.06(19.4)
Normal	5	0.56±0.14(13.9)	10	(0.00±0.1(27.8)
High	15	0.13±0.02(41.7)	6	0.41±0.26(16.6)
Cd(mg/dL)				
Normal	20	0.00±0.00(55.6)	16	0.00±0.01(44.4)

KEY: ND = Not detected NO₃⁻ = Nitrate, TDS = Total Dissolved solids, CL = Chloride, Fe = Iron E.C = Electrical Conductivity, Cu = Copper, Cd = Cadmium

Table 7 Physicochemical Properties of the Borehole Water Samples According to the Locations

Parameters	N	M±SD	Mushi (n =10)	Bariga (n =10)	Makoko (n =10)	Ajgunle (n =10)	Ijora (n =10)
PH							
Low	24	5.52±0.72	6(12)	6(12)	6(12)	2(4)	4(8)
Normal	26	7.03±0.37	4(8)	4(8)	4(8)	8(16)	6(12)
Total	50	6.30±0.94					
Temp							
Low	40	24.31±1.38	10(20)	10(20)	3(6)	10(20)	7(14)
Normal	4	26.25±0.24	-	-	2(4)	-	2(4)
High	6	27.23±0.10	-	-	5(10)	-	1(2)
Total	50	24.82±1.62					
E.Conductivity (µs/cm)							
Low	40	439.81±235.28	9(18)	9(18)	10(20)	2(4)	10(20)
High	10	1700.60±561.89	1(2)	1(2)	-	8(16)	-
Total	50	691.97±601.31					
TDS (mg/dL)							
Normal	36	244.34±137.21	8(16)	9(18)	10(20)	2(4)	7(14)
High	14	775.36±261.71	2(4)	1(2)	-	8(16)	3(6)
Total	50	393.03±299.38					
CL- (mg/dL)							
Low	47	120.89±61.45	10(20)	10(20)	10(20)	7(14)	10(20)
High	3	320.87±27.85	-	-	-	3(6)	-
Total	50	132.89±76.67					
NO ₃ -(mg/dL)							
Low	50	1.69±1.62	10(20)	10(20)	10(20)	10(20)	10(20)
Total	50	1.69±1.62					
Cu (mg/dL)							
Normal	30	0.58±0.39	3(6)	5(10)	10(20)	6(12)	6(12)
High	20	1.43±0.37	7(14)	5(10)	-	4(8)	4(8)
Total	50	0.92±0.57					
Fe (mg/dL)							
Low	25	0.13±0.08	6(12)	5(10)	3(6)	3(6)	8(16)
Normal	1	0.30±0.00	-	-	-	-	1(2)
High	24	0.98±0.64	4(8)	5(10)	7(14)	7(14)	1(2)
Total	50	0.54±0.62					

Pb (mg/dL)							
Normal	28	0.00±0.00	8(16)	4(8)	10(20)	-	6(12)
High	22	0.13±0.19	2(4)	6(12)	-	10(20)	4(8)
Total	50	0.06±0.14					
Cd(mg/dL)							
Normal	38	0.00±0.00	7(14)	7(14)	10(20)	6(12)	8(16)
High	12	0.06±0.05	3(6)	3(6)	-	4(8)	2(2)
Total	50	0.01±0.03					

KEY: ND = Not detected NO₃⁻ = Nitrate, TDS = Total Dissolved solids, CL = Chloride, Fe = Iron E.C = Electrical Conductivity, Cu = Copper, Cd = Cadmium

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